

Mehrad Haghshenas

PHD STUDENT, UNIVERSITY OF WATERLOO

Canada

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Research Interests

Formal Methods

Automated Theorem Proving

Static Analysis

Programming Languages

Software Testing

Functional Programming

Education

University of Waterloo

PH.D. IN COMPUTER SCIENCE

Canada

Expected: Sep. 2029

- Cumulative GPA: **92.00 / 100.00**.
- Member of the CrySP (Cryptography Security & Privacy) lab – supervised by Dr. Meng Xu.

Utrecht University

COMPUTING SCIENCE

Netherlands

Graduation: Jan. 2024

- Cumulative GPA: **3.94 / 4.00** (*Summa cum laude*).
- Thesis: *Automated Verification of Hoare Triples: A Comparative Study on Static Inference of Loop Invariants* – supervised by Dr. S.W.B. Wishnu Prasetya and Prof. dr. ir. Frank van der Stappen.

Awards

2025 **International Doctoral Student Award (IDSA)**, Awarded by the *University of Waterloo*.

Canada

2023 **Scholarship**, Awarded by *Utrecht University*.

Netherlands

Publications

RuSMT: An Executable Semantics as Conformance Oracle and Test Suite Synthesizer

ACM

SPECOPS 2026 (WORKSHOP, CO-LOCATED WITH SPLASH)

Oct. 2026

- **M. Haghshenas** and M. Xu. International Workshop on Specification-Driven Development Life Cycle. [doi:10.1145/3842652.3843196](https://doi.org/10.1145/3842652.3843196)

Open Source Contributions

TOML conformance suite

GitHub

OPEN-SOURCE CONTRIBUTOR TO [TOML-TEST](#).

Aug. 2026

- Found a previously unreported defect in *smol-toml*, which accepts an array-of-tables header closed by a single bracket, and [reported it upstream](#). Contributed the two missing end-of-input cases to toml-test ([PR #205](#), merged).

Rust Z3 bindings

GitHub

OPEN-SOURCE CONTRIBUTOR TO THE [Z3 RUST API](#).

Sep. 2025

- Contributed feature enhancements and correctness fixes to the bindings, and authored the [Rust bindings guide](#), published on the official Z3 project website.

Work Experience

Shopify

Canada

SOFTWARE DEVELOPER

Jan. 2026 - May 2026

- Contributed to the *agentic e-commerce* project using Ruby on Rails in the CT&I team.

ABN AMRO

Netherlands

FULL-TIME MAINFRAME DEVELOPER

Mar. 2024 - Dec. 2024

- Migrated and refactored legacy COBOL under a behavior-preserving requirement—changes had to keep the system conformant to its existing specifications. Built regression tests (*Topaz for Total Test*) to certify spec conformance across the migration, and updated technical documentation. Collaborated in Agile/Scrum teams.

Hudson Dynamics B.V.

Netherlands

EMBEDDED SOFTWARE ENGINEER INTERN

Jun. 2023 - Dec. 2023

- Worked on the development of automated vehicles. Programmed the robotic arm of an electric loader to ensure precise vertical lift path control. Designed its control system using Parker IQAN software. Modeled the arm's search space with Denavit-Hartenberg notation and applied forward and inverse kinematic analysis.

Volunteering Activities

Women in Mathematics Program

University of Waterloo

MENTOR

Winter 2026

- Designed and led a *Women in Mathematics Directed Research Program* project, mentoring undergraduate students on formally verifying SAT-solver encodings in Coq—working on a verified library that extends the FMCAD'23 *Verified Encodings* work toward cardinality, pseudo-boolean, and symmetry-breaking constraints.
- Recipient of the *Women in Mathematics Mentorship Award* for mentorship contributions.

Artifact Evaluation Committee

PoPETs

PROGRAM COMMITTEE MEMBER

Oct. 2025

- Served on the committee for the Proceedings on Privacy Enhancing Technologies symposium, reviewing the *ReportTor: Facilitating User Reporting of Issues Encountered in Naturalistic Web Browsing via the Tor Browser* artifact to ensure reproducibility.
- Contributed to the open science mission by providing feedback to authors on artifact quality, completeness, and reusability.

Hack the North

University of Waterloo

PARTICIPANT

12 – 14 Sep. 2025

- We developed *CodeMind*, a local-first, searchable timeline tool that documents AI coding sessions. The tool uses a custom Model Context Protocol (MCP) server to ingest sessions from Cursor/Claude, and an OpenAI-based LLM to extract and index key information.

Stanford University Code in Place

Online

VOLUNTEER TUTOR

Apr.–May 2023, 2024, & 2025

- Volunteer tutor for Stanford's Code in Place (2023, 2024, 2025), mentoring 15 students over 6 weeks each year.

Summer School

Queen Mary University of London

England

THREE-WEEK COURSE: *Business and Society: The Changing World of Work*

Jul. 2022–Aug. 2022

- 15 UK credits (7.5 ECTS); grade: **A** (75.6/100).

Utrecht University

Netherlands

ONE-WEEK COURSE ON *Advanced Functional Programming in Haskell* (DR. WOUTER SWIERSTRA)

Jul. 2022

- No exam was administered.

Miscellaneous

Languages	English (IELTS 8.5 , taken on Nov. 3, 2023), Persian (native), French (A1).
Programming	Rust, C, Python, Haskell, Java, COBOL, MATLAB, JavaScript, R, SQL, C#.
Verification Tools	Coq (proof assistant), Z3 (SMT), SPIN (model checker).
Relevant Courses	Software & Systems Security, Languages & Compilers, Functional Programming, Software Testing & Verification, Logic for Computer Science, Operating Systems, Software Verification Using Proof Assistants, Machine Learning.